

AMR / ECSA / FFG Master Dossier

AMR, AI-for-science and project execution.

ECSA Predictive AMR AI Architecture

I wanted a serious applied target for uncertainty-aware AI: antimicrobial resistance modeled as a probabilistic graph/state-space system rather than as a static dashboard or one-shot black-box prediction.

- This is the best future-project answer: a concrete AI-for-science direction where uncertainty, validation, lab feedback and governance matter.
- Architecture and project direction, not clinically validated software.

What I had in mind

ECSA is the project where I would like to turn systems thinking toward a serious scientific problem. The core idea is not to make a medical claim, but to design an uncertainty-aware proof of concept that can start safely on synthetic and public data.

What I built

- Detailed architecture proposal for AMR dynamics as probabilistic graph/state-space modeling.
- Resistance, uncertainty and stability parameters with lab-in-the-loop updates.
- Bayesian inference, graph learning, optional RL path search, regulatory/open-science governance.

Mechanism detail

- ECSA models antimicrobial resistance as a dynamic probabilistic graph/state-space system rather than a static dashboard.
- The important parameters are resistance, uncertainty and stability. The design can integrate Bayesian updates, graph learning, optional path-search/RL ideas and lab-in-the-loop validation.
- Its safest first implementation is synthetic/public data, not clinical use.

Architecture

- Wet-lab or public/synthetic input data.
- Probabilistic resistance graph with R, U/sigma and S parameters.
- Bayesian update layer and graph learning layer.
- Hypotheses returned to validation loop.
- Governance, audit, intended-use and privacy layer.

Evidence cluster

- AMR proposal text export documents probabilistic model, FAIR corpus, lab-in-the-loop validation and governance.
- PCT-style drafting describes a computer-implemented dynamic probabilistic state-space system.
- Utility-model text describes graph-based adaptive resistance modeling.

Claim-to-evidence map

The public version should be strong because it is bounded, not because it overclaims.

Claim	Evidence	Boundary
This is the best future-project answer: a concrete AI-for-science direction where uncertainty, validation, lab feedback and governance matter.	AMR proposal text export documents probabilistic model, FAIR corpus, lab-in-the-loop validation and governance.; PCT-style drafting describes a computer-implemented dynamic probabilistic state-space system.	Architecture and project direction, not clinically validated software.
The work is valuable because it shows mechanism, not surface polish.	Wet-lab or public/synthetic input data.; Probabilistic resistance graph with R, U/sigma and S parameters.	Fresh public demos or benchmarks would be the next proof layer.

Senior-level signal

This is the best future-project answer: a concrete AI-for-science direction where uncertainty, validation, lab feedback and governance matter.

- High-stakes uncertainty: outputs should carry confidence and boundary, not false certainty.
- AI-for-science staging: simulator first, then update layer, then external review.
- Governance-first thinking: data provenance, model cards and intended-use boundaries before real-world claims.

Skeptical reviewer questions

- What is actually built here, and what is still design? Answer: the status and boundary are stated directly inside the ECSA Predictive AMR AI Architecture case.
- Which private evidence is intentionally not public? Answer: local paths, credentials, private channel details and confidential raw material are excluded.
- Why is this relevant? Answer: the case shows a reusable component for agents, tool use, memory, routing, validation or high-stakes governance.
- What would be the next stronger proof? Answer: synthetic demo, audit trace, benchmark or external review, depending on the case.

Lessons and boundaries

Architecture and project direction, not clinically validated software.

- High-stakes AI outputs must carry uncertainty rather than pretend certainty.
- The safe starting point is synthetic/public data and proof-of-concept staging.
- Governance has to be designed before real-world medical claims.

Next hardening / next project step

- Build a synthetic AMR simulator.
- Add Bayesian update and graph learning experiments.
- Create model cards, data provenance, audit logs and intended-use boundaries.
- Seek external scientific/lab review before clinical claims.
- Build a synthetic AMR simulator with R/U/S parameters.

- Evaluate Bayesian updates and graph link prediction on withheld synthetic relationships.
- Create model cards, audit logs and human-review gates before any real data claim.

AMR-HORIZONT / FFG Impact Innovation Package

I wanted to convert AMR from a technical idea into a participatory One-Health project format with stakeholders, outputs, work packages and measurable validation logic.

- This shows that I can organize beyond code: work packages, budget logic, stakeholder strategy, evaluator Q&A, red flags and scenario tests.
- Prepared/submitted project package, not proof of funded execution unless separately confirmed.

What I had in mind

AMR-HORIZONT shows a different side of the work: not just algorithmic architecture, but project structure, stakeholders, budget logic, risks, outputs and evaluator-facing answers.

What I built

- 12-month project plan, work packages, budget logic and stakeholder map.
- Evaluator Q&A, red-flag handling, LOI/support strategy and scenario-based AP5 testing.

Mechanism detail

- AMR-HORIZONT translates a technical AMR idea into a participatory One-Health project package.
- The evidence is project-management depth: work packages, budget logic, stakeholder groups, output maps, evaluator Q&A, red flags, support strategy and scenario-based AP5 testing.
- It should be presented as prepared/submitted planning evidence unless separate funding approval is shown.

Architecture

- Problem framing and Theory of Change.
- Stakeholder interviews and needs matrix.
- Participatory decision framework.
- Toolkit, policy brief, pilot report and utilization roadmap.

Evidence cluster

- FFG project-management evidence note documents work packages, outputs, Q&A and risk mitigation.
- Use as planning/submission evidence; funding outcome is separate.

Claim-to-evidence map

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Claim	Evidence	Boundary
This shows that I can organize beyond code: work packages, budget logic, stakeholder strategy, evaluator Q&A, red flags and scenario tests.	FFG project-management evidence note documents work packages, outputs, Q&A and risk mitigation.; Use as planning/submission evidence; funding outcome is separate.	Prepared/submitted project package, not proof of funded execution unless separately confirmed.

Claim	Evidence	Boundary
The work is valuable because it shows mechanism, not surface polish.	Problem framing and Theory of Change.; Stakeholder interviews and needs matrix.	Fresh public demos or benchmarks would be the next proof layer.

Senior-level signal

This shows that I can organize beyond code: work packages, budget logic, stakeholder strategy, evaluator Q&A, red flags and scenario tests.

- Complex project organization beyond code.
- Stakeholder and governance thinking in a high-stakes domain.
- Ability to anticipate evaluator objections before the conversation.

Skeptical reviewer questions

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- Why is this relevant? Answer: the case shows a reusable component for agents, tool use, memory, routing, validation or high-stakes governance.
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Lessons and boundaries

Prepared/submitted project package, not proof of funded execution unless separately confirmed.

- A strong technical idea also needs stakeholder and governance design.
- Budget, outputs, risks and evaluation need to be thought together.

Next hardening / next project step

- Use as general-portfolio proof of project maturity.
- Keep secondary in Fellowship unless the project question needs it.
- Create a public sanitized project one-pager with work packages, outputs and risk table.
- Keep private budget/personal details out of the public package unless intentionally disclosed.